

Algorithms (Algo)

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 259

Question 1 of 259 | GATE 2025 Set 1 (CS1) | Q.20 (Computer Science & IT)

Q.20

Consider the following recurrence relation:

$$T(n) = 2T(n-1) + n2^n \text{ for } n > 0, T(0) = 1.$$

Which ONE of the following options is CORRECT?

- (A) $T(n) = \Theta(n2^{2n})$
- (B) $T(n) = \Theta(n2^n)$
- (C) $T(n) = \Theta((\log n)^2 2^n)$
- (D) $T(n) = \Theta(4^n)$

Question 2 of 259 | GATE 2025 Set 1 (CS1) | Q.25 (Computer Science & IT)

Q.25

The number -6 can be represented as 1010 in 4-bit 2's complement representation.

Which of the following is/are CORRECT 2's complement representation(s) of -6 ?

- (A) 1000 1010 in 8-bits
- (B) 1111 1010 in 8-bits
- (C) 1000 0000 0000 1010 in 16-bits
- (D) 1111 1111 1111 1010 in 16-bits

Question 3 of 259 | GATE 2025 Set 1 (CS1) | Q.33 (Computer Science & IT)

Q.33

The pseudocode of a function `fun()` is given below:

```
fun(int A[0,...,n-1]){
  for i=0 to n-2
    for j=0 to n-i-2
      if (A[j]>A[j+1])
        then swap A[j] and A[j+1]
    }
```

Let $A[0, \dots, 29]$ be an array storing 30 distinct integers in descending order. The number of swap operations that will be performed, if the function `fun()` is called with $A[0, \dots, 29]$ as argument, is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 4 of 259 | GATE 2025 Set 2 (CS2) | Q.11 (Computer Science & IT)

Q.11

If $A = \begin{pmatrix} 1 \\ 2 \\ 2 \\ -1 \end{pmatrix}$,

2

2

then which ONE of the following is A^8 ?

- (A) $\begin{pmatrix} 25 & 0 & 0 & 25 \end{pmatrix}$
- (B) $\begin{pmatrix} 125 & 0 & 0 & 125 \end{pmatrix}$
- (C) $\begin{pmatrix} 625 & 0 & 0 & 625 \end{pmatrix}$
- (D) $\begin{pmatrix} 3125 & 0 & 0 & 3125 \end{pmatrix}$

Question 5 of 259 | GATE 2025 Set 2 (CS2) | Q.15 (Computer Science & IT)

Q.15

Let $P(x)$ be an arbitrary predicate over the domain of natural numbers.

Which ONE of the following statements is TRUE?

- (A) $(P(0) \wedge (\forall x [P(x) \wedge P(x+1)])) \wedge (\forall x P(x))$
- (B) $(P(0) \wedge (\forall x [P(x) \wedge P(x-1)])) \wedge (\forall x P(x))$
- (C) $(P(1000) \wedge (\forall x [P(x) \wedge P(x-1)])) \wedge (\forall x P(x))$
- (D) $(P(1000) \wedge (\forall x [P(x) \wedge P(x+1)])) \wedge (\forall x P(x))$

Question 6 of 259 | GATE 2025 Set 2 (CS2) | Q.20 (Computer Science & IT)

Q.20

Consider an unordered list of N distinct integers.

What is the minimum number of element comparisons required to find an integer in the list that is NOT the largest in the list?

- (A) 1
- (B) $N-1$
- (C) N
- (D) $2N-1$

Question 7 of 259 | GATE 2025 Set 2 (CS2) | Q.29 (Computer Science & IT)

Q.29

Which of the following statements regarding Breadth First Search (BFS) and Depth First Search (DFS) on an undirected simple graph G is/are TRUE?

- (A) A DFS tree of G is a Shortest Path tree of G .

- (B) Every non-tree edge of G with respect to a DFS tree is a forward/back edge.
- (C) If (u, v) is a non-tree edge of G with respect to a BFS tree, then the distances from the source vertex s to u and v in the BFS tree are within ± 1 of each other.
- (D) Both BFS and DFS can be used to find the connected components of G .

Question 8 of 259 | GATE 2025 Set 2 (CS2) | Q.31 (Computer Science & IT)

Q.31

Consider the following logic circuit diagram.

Which is/are the CORRECT option(s) for the output function F ?

- (A) $X \bar{Y}$
- (B) $X + \bar{Y} + XY$
- (C) $X \bar{Y} + X + XY$
- (D) $X + \bar{Y}$

Question 9 of 259 | GATE 2025 Set 2 (CS2) | Q.50 (Computer Science & IT)

Q.50

Which of the following Boolean algebraic equation(s) is/are CORRECT?

- (A) $\bar{A} \bar{B} C + A \bar{B} \bar{C} + \bar{A} \bar{B} \bar{C} + A \bar{B} C + A B C = B C + \bar{B} \bar{C} + \bar{A} \bar{B}$
- (B) $A B + \bar{A} C + B C = A B + \bar{A} C$
- (C) $(A + C)(\bar{A} + B) = A B + \bar{A} C$
- (D) $(A + \bar{B} + \bar{D})(C + D)(\bar{A} + C + D)(A + B + \bar{D}) = \bar{A} \bar{D} + C \bar{D}$

Question 10 of 259 | GATE 2025 Set 2 (CS2) | Q.59 (Computer Science & IT)

Q.59

Consider the following algorithm someAlgo that takes an undirected graph G as input.

someAlgo(G)

1. Let v be any vertex in G . Run BFS on G starting at v . Let u be a vertex in G at maximum distance from v as given by the BFS.
2. Run BFS on G again with u as the starting vertex. Let z be the vertex at maximum distance from u as given by the BFS.
3. Output the distance between u and z in G .

The output of someAlgo(T) for the tree shown in the given figure is _____. (Answer in integer)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 11 of 259 | GATE 2025 Set 2 (CS2) | Q.65 (Computer Science & IT)

Q.65

The unit interval $(0,1)$ is divided at a point chosen uniformly distributed over $(0,1)$ in $\mathbb{R}$ into two disjoint subintervals.

The expected length of the subinterval that contains 0.4 is _____. (rounded off to two decimal places)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 12 of 259 | GATE 2024 Set 1 (CS1 S5) | Q.13 (Computer Science & IT)

Q.13

Consider a system that uses 5 bits for representing signed integers in 2's complement format. In this system, two integers A and B are represented as $A=01010$ and $B=11010$. Which one of the following operations will result in either an arithmetic overflow or an arithmetic underflow?

- (A) $A + B$
- (B) $A - B$
- (C) $B - A$
- (D) $2 \oplus B$

Question 13 of 259 | GATE 2024 Set 1 (CS1 S5) | Q.34 (Computer Science & IT)

Q.34

The number of spanning trees in a complete graph of 4 vertices labelled A, B, C, and D is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 14 of 259 | GATE 2024 Set 1 (CS1 S5) | Q.42 (Computer Science & IT)

Q.42

Consider the following recurrence relation:

$$T(n) = \sqrt{n}T(\sqrt{n}) + n \text{ for } n \geq 1,$$

1 for $n = 1$.

Which one of the following options is CORRECT?

- (A) $T(n) = \Theta(n \log \log n)$
- (B) $T(n) = \Theta(n \log n)$
- (C) $T(n) = \Theta(n^2 \log n)$

(D) $T(n) = \Theta(n^2 \log \log n)$

Question 15 of 259 | GATE 2024 Set 1 (CS1 S5) | Q.54 (Computer Science & IT)

Q.54

Consider a 512 GB hard disk with 32 storage surfaces. There are 4096 sectors per track and each sector holds 1024 bytes of data. The number of cylinders in the hard disk is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 16 of 259 | GATE 2024 Set 1 (CS1 S5) | Q.55 (Computer Science & IT)

Q.55

The baseline execution time of a program on a 2 GHz single core machine is 100 nanoseconds (ns). The code corresponding to 90% of the execution time can be fully parallelized. The overhead for using an additional core is 10 ns when running on a multicore system. Assume that all cores in the multicore system run their share of the parallelized code for an equal amount of time.

The number of cores that minimize the execution time of the program is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 17 of 259 | GATE 2024 Set 1 (CS1 S5) | Q.62 (Computer Science & IT)

Q.62

Consider a memory management system that uses a page size of 2 KB. Assume that both the physical and virtual addresses start from 0. Assume that the pages 0, 1, 2, and 3 are stored in the page frames 1, 3, 2, and 0, respectively. The physical address (in decimal format) corresponding to the virtual address 2500 (in decimal format) is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 18 of 259 | GATE 2024 Set 1 (CS1 S5) | Q.65 (Computer Science & IT)

Q.65

Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 19 of 259 | GATE 2024 Set 2 (CS2 S6) | Q.15 (Computer Science & IT)

Q.15

Let $T(n)$ be the recurrence relation defined as follows:

$$T(0) = 1,$$

$$T(1) = 2, \text{ and}$$

$$T(n) = 5T(n-1) - 6T(n-2) \text{ for } n \geq 2$$

Which one of the following statements is TRUE?

- (A) $T(n) = \Theta(2^n)$
- (B) $T(n) = \Theta(n^2)$
- (C) $T(n) = \Theta(3^n)$
- (D) $T(n) = \Theta(n^3)$

Question 20 of 259 | GATE 2024 Set 2 (CS2 S6) | Q.19 (Computer Science & IT)

Q.19

Once the DBMS informs the user that a transaction has been successfully completed, its effect should persist even if the system crashes before all its changes are reflected on disk. This property is called

- (A) durability
- (B) atomicity
- (C) consistency
- (D) isolation

Question 21 of 259 | GATE 2024 Set 2 (CS2 S6) | Q.53 (Computer Science & IT)

Q.53

Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 sectors/track, 512-byte sectors. A file has content stored in 3000 sectors located randomly on the disk. Assuming average rotational latency, the total time (in seconds, rounded off to 2 decimal places) to read the entire file from the disk is _____

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 22 of 259 | GATE 2024 Set 2 (CS2 S6) | Q.59 (Computer Science & IT)

Q.59

The number of distinct minimum-weight spanning trees of the following graph is _____

- a
- b
- g

d
c
f
e
3
2
2
1
2
3
2
2
2
1
1
1

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 23 of 259 | GATE 2023 Single Session | Q.14 (Computer Science & IT)

Q.14

The Lucas sequence L_n is defined by the recurrence relation:

$$L_n = L_{n-1} + L_{n-2},$$

for

$$n \geq 3,$$

with $L_1 = 1$ and $L_2 = 3$.

Which one of the options given is TRUE?

- (A) $L_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n + \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$
- (B) $L_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$
- (C) $L_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n + \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$
- (D) $L_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$ CS

Question 24 of 259 | GATE 2023 Single Session | Q.25 (Computer Science & IT)

Q.25

Geetha has a conjecture about integers, which is of the form

$$\forall x$$

$$P(x) \Rightarrow \exists y Q(x, y)$$

,

where P is a statement about integers, and Q is a statement about pairs of integers.

Which of the following (one or more) option(s) would imply Geetha's conjecture?

- (A) $\exists x P(x) \wedge \forall y Q(x, y)$
- (B) $\exists x \forall y Q(x, y)$
- (C) $\forall y \exists x P(x) \Rightarrow Q(x, y)$
- (D) $\exists x P(x) \wedge \forall y Q(x, y)$ CS

Question 25 of 259 | GATE 2023 Single Session | Q.28 (Computer Science & IT)

Q.28

Let f and g be functions of natural numbers given by $f(n) = n$ and $g(n) = n^2$.

Which of the following statements is/are TRUE?

- (A) $f \in O(g)$
- (B) $f \in \Omega(g)$
- (C) $f \in o(g)$
- (D) $f \in \Theta(g)$ CS

Question 26 of 259 | GATE 2023 Single Session | Q.31 (Computer Science & IT)

Q.31

A particular number is written as 132 in radix-4 representation. The same number in radix-5 representation is

.

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 27 of 259 | GATE 2023 Single Session | Q.35 (Computer Science & IT)

Q.35

CS

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 28 of 259 | GATE 2023 Single Session | Q.50 (Computer Science & IT)

Q.50

Let X be a set and $2X$ denote the powerset of X .

Define a binary operation Δ on $2X$ as follows:

$$A \Delta B = (A - B) \cup (B - A).$$

Let $H = (2X, \Delta)$. Which of the following statements about H is/are correct?

- (A) H is a group.
- (B) Every element in H has an inverse, but H is NOT a group.

- (C) For every $A \subseteq X$, the inverse of A is the complement of A .
- (D) For every $A \subseteq X$, the inverse of A is A . CS

Question 29 of 259 | GATE 2023 Single Session | Q.53 (Computer Science & IT)

Q.53

Consider functions Function 1 and Function 2 expressed in pseudocode as follows:

Function 1

```
while n > 1 do
  for i = 1 to n do
    x = x + 1;
  end for
  n = ⌊n/2⌋;
end while
```

Function 2

```
for i = 1 to 100 ⌊n⌋ do
  x = x + 1;
end for
```

Let $f_1(n)$ and $f_2(n)$ denote the number of times the statement " $x = x + 1$ " is executed in Function 1 and Function 2, respectively.

Which of the following statements is/are TRUE?

- (A) $f_1(n) \in O(f_2(n))$
- (B) $f_1(n) \in o(f_2(n))$
- (C) $f_1(n) \in \omega(f_2(n))$
- (D) $f_1(n) \in O(n)$ CS

Question 30 of 259 | GATE 2023 Single Session | Q.65 (Computer Science & IT)

Q.65

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 31 of 259 | GATE 2022 Single Session | Q.27 (Computer Science & IT)

Q.27

Which of the following statements is/are TRUE for a group

?

G

- (A) If for all $x, y \in G$, $x^2 y^2 = (xy)^2$ then G is commutative.
- (B) If for all $x \in G$, $x^2 = 1$, $x \neq 1$ then G is commutative. Here, 1 is the identity element of G .

- (C) If the order of G is 2, then G is commutative.
- (D) If G is commutative, then a subgroup of G need not be commutative.

Question 32 of 259 | GATE 2022 Single Session | Q.30 (Computer Science & IT)

Q.30

Consider a simple undirected graph of 10 vertices. If the graph is disconnected, then the maximum number of edges it can have is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 33 of 259 | GATE 2022 Single Session | Q.32 (Computer Science & IT)

Q.32

The number of arrangements of six identical balls in three identical bins is_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 34 of 259 | GATE 2022 Single Session | Q.39 (Computer Science & IT)

Q.39

Let

()

iR z and

()

i

$W z$ denote read and write operations on a data element z by a transaction

iT , respectively. Consider the schedule S with four transactions.

S:

4

2

3

1

1

2

3

4

()

()

()

()

()

()

()

()

$$R_x R_x R_x R_y W_y W_x W_y R_y$$

Which one of the following serial schedules is conflict equivalent to S?

- (A) $TQ \rightarrow TQ \rightarrow TQ \rightarrow TQ$
- (B) $TQ \rightarrow TQ \rightarrow TQ \rightarrow TQ$
- (C) $TQ \rightarrow TQ \rightarrow TQ \rightarrow TQ$
- (D) $TQ \rightarrow TQ \rightarrow TQ \rightarrow TQ$

Question 35 of 259 | GATE 2022 Single Session | Q.51 (Computer Science & IT)

Q.51

Consider the following recurrence:

(1)

1;

(2)

2 ()

1,

for

1;

(2

1)

2 ()

1,

for

1.

f

f

n

 f_n

n

f

n

 f_n

n

1

1

1

1

1

1

1

1

Then, which of the following statements is/are TRUE?

- (A) $(2 \ 1) \ 2 \ 1 \ n \ n \ f \ \square \ \square \ \square$
- (B) $(2) \ 1 \ n \ f \ \square$
- (C) $1 \ (5 \ 2) \ 2 \ 1 \ n \ n \ f \ \square \ \square \ \square \ \square$
- (D) $(2 \ 1) \ 2 \ 1 \ n \ n \ f \ \square \ \square \ \square$

Question 36 of 259 | GATE 2021 Set 1 (CS1) | Q.2 (Computer Science & IT)

Q.2

A polygon is convex if, for every pair of points, P and Q belonging to the polygon, the line segment PQ lies completely inside or on the polygon.

Which one of the following is NOT a convex polygon?

- (A)
- (B)
- (C)
- (D)

Question 37 of 259 | GATE 2021 Set 1 (CS1) | Q.3 (Computer Science & IT)

Q.3

Consider the following sentences:

(i)

Everybody in the class is prepared for the exam.

(ii)

Babu invited Danish to his home because he enjoys playing chess.

Which of the following is the CORRECT observation about the above two sentences?

- (A) (i) is grammatically correct and (ii) is unambiguous
- (B) (i) is grammatically incorrect and (ii) is unambiguous
- (C) (i) is grammatically correct and (ii) is ambiguous
- (D) (i) is grammatically incorrect and (ii) is ambiguous

Question 38 of 259 | GATE 2021 Set 1 (CS1) | Q.7 (Computer Science & IT)

Q.7

Items

Cost

(₹)

Profit %

Marked Price

(₹)

P

5,400

5,860

Q

25

10,000

Details of prices of two items P and Q are presented in the above table. The ratio of cost of item P to cost of item Q is 3:4. Discount is calculated as the difference between the marked price and the selling price. The profit percentage is calculated as the ratio of the difference between selling price and cost, to the cost (Profit % =

$\frac{\text{Selling price} - \text{Cost}}{\text{Cost}}$

$\times 100$).

The discount on item Q, as a percentage of its marked price, is _____

- (A) 25
- (B) 12.5
- (C) 10
- (D) 5

Question 39 of 259 | GATE 2021 Set 1 (CS1) | Q.9 (Computer Science & IT)

Q.9

Given below are two statements 1 and 2, and two conclusions I and II.

Statement 1: All bacteria are microorganisms.

Statement 2: All pathogens are microorganisms.

Conclusion I: Some pathogens are bacteria.

Conclusion II: All pathogens are not bacteria.

Based on the above statements and conclusions, which one of the following options is logically CORRECT?

- (A) Only conclusion I is correct
- (B) Only conclusion II is correct
- (C) Either conclusion I or II is correct.
- (D) Neither conclusion I nor II is correct.

Question 40 of 259 | GATE 2021 Set 1 (CS1) | Q.1 (Computer Science & IT)

Q.11 – Q.15 Multiple Select Question (MSQ), carry ONE mark each (no negative marks).

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 41 of 259 | GATE 2021 Set 1 (CS1) | Q.1 (Computer Science & IT)

Q.16 – Q.25 Numerical Answer Type (NAT), carry ONE mark each (no negative marks).

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 42 of 259 | GATE 2021 Set 1 (CS1) | Q.2 (Computer Science & IT)

Q.26 – Q.39 Multiple Choice Question (MCQ), carry TWO mark each (for each wrong answer: – 2/3).

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 43 of 259 | GATE 2021 Set 1 (CS1) | Q.4 (Computer Science & IT)

Q.40 – Q.47 Multiple Select Question (MSQ), carry TWO mark each (no negative marks).

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 44 of 259 | GATE 2021 Set 1 (CS1) | Q.4 (Computer Science & IT)

Q.48 – Q.55 Numerical Answer Type (NAT), carry TWO mark each (no negative marks).

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 45 of 259 | GATE 2021 Set 2 (CS2) | Q.1 (Computer Science & IT)

Q.1

Gauri said that she can play the keyboard _____ her sister.

- (A) as well as
- (B) as better as
- (C) as nicest as
- (D) as worse as

Question 46 of 259 | GATE 2021 Set 2 (CS2) | Q.5 (Computer Science & IT)

Q.5

Pen : Write :: Knife : _____

Which one of the following options maintains a similar logical relation in the above?

- (A) Vegetables
- (B) Sharp

(C) Cut

(D) Blunt Q. 6 – Q. 10 Multiple Choice Question (MCQ), carry TWO marks each (for each wrong answer: – 2/3).

Question 47 of 259 | GATE 2021 Set 2 (CS2) | Q.7 (Computer Science & IT)

Q.7

A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.

(A)

(B)

(C)

(D)

Question 48 of 259 | GATE 2021 Set 2 (CS2) | Q.8 (Computer Science & IT)

Q.8

The number of students in three classes is in the ratio 3:13:6. If 18 students are added to each class, the ratio changes to 15:35:21.

The total number of students in all the three classes in the beginning was:

(A) 22

(B) 66

(C) 88

(D) 110

Question 49 of 259 | GATE 2021 Set 2 (CS2) | Q.9 (Computer Science & IT)

Q.9

The number of units of a product sold in three different years and the respective net profits are presented in the figure above. The cost/unit in Year 3 was ₹ 1, which was half the cost/unit in Year 2. The cost/unit in Year 3 was one-third of the cost/unit in Year 1. Taxes were paid on the selling price at 10%, 13% and 15% respectively for the three years. Net profit is calculated as the difference between the selling price and the sum of cost and taxes paid in that year.

The ratio of the selling price in Year 2 to the selling price in Year 3 is _____.

A) 4:3

(B) 1:1

(C) 3:4

- (D) 1:2 100 200 300 240 296 210 0 50 100 150 200 250 300 350 Year 1 Year 2 Year 3 Number of units Net Profit (INR) (₹)

Question 50 of 259 | GATE 2021 Set 2 (CS2) | Q.10 (Computer Science & IT)

Q.10

Six students P, Q, R, S, T and U, with distinct heights, compare their heights and make the following observations.

Observation I: S is taller than R.

Observation II: Q is the shortest of all.

Observation III: U is taller than only one student.

Observation IV: T is taller than S but is not the tallest.

The number of students that are taller than R is the same as the number of students shorter than ____.

- (A) T
- (B) R
- (C) S
- (D) P Q.1 – Q.10 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

Question 51 of 259 | GATE 2021 Set 2 (CS2) | Q.1 (Computer Science & IT)

Q.11 – Q.15 Multiple Select Question (MSQ), carry ONE mark each (no negative marks).

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 52 of 259 | GATE 2021 Set 2 (CS2) | Q.1 (Computer Science & IT)

Q.16 – Q.25 Numerical Answer Type (NAT), carry ONE mark each (no negative marks).

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 53 of 259 | GATE 2021 Set 2 (CS2) | Q.2 (Computer Science & IT)

Q.26 – Q.39 Multiple Choice Question (MCQ), carry TWO mark each (for each wrong answer: – 2/3).

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 54 of 259 | GATE 2021 Set 2 (CS2) | Q.4 (Computer Science & IT)

Q.40 – Q.47 Multiple Select Question (MSQ), carry TWO mark each (no negative marks).

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 55 of 259 | GATE 2021 Set 2 (CS2) | Q.4 (Computer Science & IT)

Q.48 – Q.55 Numerical Answer Type (NAT), carry TWO mark each (no negative marks).

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 56 of 259 | GATE 2020 Single Session | Q.1 (Computer Science & IT)

Q.No. 1

(A)

(B)

(C)

(D)

Question 57 of 259 | GATE 2020 Single Session | Q.2 (Computer Science & IT)

Q.No. 2

(A)

(B)

(C)

(D)

Question 58 of 259 | GATE 2020 Single Session | Q.3 (Computer Science & IT)

Q.No. 3

(A)

(B)

(C)

(D)

Question 59 of 259 | GATE 2020 Single Session | Q.4 (Computer Science & IT)

Q.No. 4

(A)

(B)

(C)

(D)

Question 60 of 259 | GATE 2020 Single Session | Q.5 (Computer Science & IT)

Q.No. 5

(A)

(B)

(C)

(D)

Question 61 of 259 | GATE 2020 Single Session | Q.6 (Computer Science & IT)

Q.No. 6

(A)

(B)

(C)

(D)

Question 62 of 259 | GATE 2020 Single Session | Q.7 (Computer Science & IT)

Q.No. 7

(A)

(B)

(C)

(D)

Question 63 of 259 | GATE 2020 Single Session | Q.8 (Computer Science & IT)

Q.No. 8

(A)

(B)

(C)

(D)

Question 64 of 259 | GATE 2020 Single Session | Q.9 (Computer Science & IT)

Q.No. 9

(A)

(B)

(C)

(D)

Question 65 of 259 | GATE 2020 Single Session | Q.10 (Computer Science & IT)

Q.No. 10

- (A)
- (B)
- (C)
- (D)

Question 66 of 259 | GATE 2020 Single Session | Q.11 (Computer Science & IT)

Q.No. 11

- (A)
- (B)
- (C)
- (D)

Question 67 of 259 | GATE 2020 Single Session | Q.12 (Computer Science & IT)

Q.No. 12

- (A)
- (B)
- (C)
- (D)

Question 68 of 259 | GATE 2020 Single Session | Q.13 (Computer Science & IT)

Q.No. 13

- (A)
- (B)
- (C)
- (D)

Question 69 of 259 | GATE 2020 Single Session | Q.14 (Computer Science & IT)

Q.No. 14

- (A)

(B)

(C)

(D)

Question 70 of 259 | GATE 2020 Single Session | Q.15 (Computer Science & IT)

Q.No. 15

(A)

(B)

(C)

(D)

Question 71 of 259 | GATE 2020 Single Session | Q.16 (Computer Science & IT)

Q.No. 16

(A)

(B)

(C)

(D)

Question 72 of 259 | GATE 2020 Single Session | Q.17 (Computer Science & IT)

Q.No. 17

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 73 of 259 | GATE 2020 Single Session | Q.18 (Computer Science & IT)**

Q.No. 18

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 74 of 259 | GATE 2020 Single Session | Q.19 (Computer Science & IT)**

Q.No. 19

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 75 of 259 | GATE 2020 Single Session | Q.20 (Computer Science & IT)**

Q.No. 20

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 76 of 259 | GATE 2020 Single Session | Q.21 (Computer Science & IT)

Q.No. 21

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 77 of 259 | GATE 2020 Single Session | Q.22 (Computer Science & IT)**

Q.No. 22

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 78 of 259 | GATE 2020 Single Session | Q.23 (Computer Science & IT)**

Q.No. 23

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 79 of 259 | GATE 2020 Single Session | Q.24 (Computer Science & IT)**

Q.No. 24

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 80 of 259 | GATE 2020 Single Session | Q.25 (Computer Science & IT)**

Q.No. 25

Q26 - Q55 carry two marks each.

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 81 of 259 | GATE 2020 Single Session | Q.26 (Computer Science & IT)**

Q.No. 26

(A)

(B)

(C)

(D)

Question 82 of 259 | GATE 2020 Single Session | Q.27 (Computer Science & IT)

Q.No. 27

(A)

(B)

(C)

(D)

Question 83 of 259 | GATE 2020 Single Session | Q.28 (Computer Science & IT)

Q.No. 28

- (A)
- (B)
- (C)
- (D)

Question 84 of 259 | GATE 2020 Single Session | Q.29 (Computer Science & IT)

Q.No. 29

- (A)
- (B)
- (C)
- (D)

Question 85 of 259 | GATE 2020 Single Session | Q.30 (Computer Science & IT)

Q.No. 30

- (A)
- (B)
- (C)
- (D)

Question 86 of 259 | GATE 2020 Single Session | Q.31 (Computer Science & IT)

Q.No. 31

- (A)
- (B)
- (C)
- (D)

Question 87 of 259 | GATE 2020 Single Session | Q.32 (Computer Science & IT)

Q.No. 32

- (A)

(B)

(C)

(D)

Question 88 of 259 | GATE 2020 Single Session | Q.33 (Computer Science & IT)

Q.No. 33

(A)

(B)

(C)

(D)

Question 89 of 259 | GATE 2020 Single Session | Q.34 (Computer Science & IT)

Q.No. 34

(A)

(B)

(C)

(D)

Question 90 of 259 | GATE 2020 Single Session | Q.35 (Computer Science & IT)

Q.No. 35

(A)

(B)

(C)

(D)

Question 91 of 259 | GATE 2020 Single Session | Q.36 (Computer Science & IT)

Q.No. 36

(A)

(B)

(C)

(D)

Question 92 of 259 | GATE 2020 Single Session | Q.37 (Computer Science & IT)

Q.No. 37

- (A)
- (B)
- (C)
- (D)

Question 93 of 259 | GATE 2020 Single Session | Q.38 (Computer Science & IT)

Q.No. 38

- (A)
- (B)
- (C)
- (D)

Question 94 of 259 | GATE 2020 Single Session | Q.39 (Computer Science & IT)

Q.No. 39

- (A)
- (B)
- (C)
- (D)

Question 95 of 259 | GATE 2020 Single Session | Q.40 (Computer Science & IT)

Q.No. 40

- (A)
- (B)
- (C)
- (D)

Question 96 of 259 | GATE 2020 Single Session | Q.41 (Computer Science & IT)

Q.No. 41

- (A)
- (B)

(C)

(D)

Question 97 of 259 | GATE 2020 Single Session | Q.42 (Computer Science & IT)

Q.No. 42

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 98 of 259 | GATE 2020 Single Session | Q.43 (Computer Science & IT)**

Q.No. 43

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 99 of 259 | GATE 2020 Single Session | Q.44 (Computer Science & IT)**

Q.No. 44

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 100 of 259 | GATE 2020 Single Session | Q.45 (Computer Science & IT)**

Q.No. 45

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 101 of 259 | GATE 2020 Single Session | Q.46 (Computer Science & IT)**

Q.No. 46

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 102 of 259 | GATE 2020 Single Session | Q.47 (Computer Science & IT)**

Q.No. 47

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 103 of 259 | GATE 2020 Single Session | Q.48 (Computer Science & IT)**

Q.No. 48

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 104 of 259 | GATE 2020 Single Session | Q.49 (Computer Science & IT)**

Q.No. 49

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 105 of 259 | GATE 2020 Single Session | Q.50 (Computer Science & IT)

Q.No. 50

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 106 of 259 | GATE 2020 Single Session | Q.51 (Computer Science & IT)**

Q.No. 51

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 107 of 259 | GATE 2020 Single Session | Q.52 (Computer Science & IT)**

Q.No. 52

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 108 of 259 | GATE 2020 Single Session | Q.53 (Computer Science & IT)**

Q.No. 53

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 109 of 259 | GATE 2020 Single Session | Q.54 (Computer Science & IT)**

Q.No. 54

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 110 of 259 | GATE 2020 Single Session | Q.55 (Computer Science & IT)**

Q.No. 55

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*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 111 of 259 | GATE 2019 Single Session | Q.2 (Computer Science & IT)**

Q.2

The chip select logic for a certain DRAM chip in a memory system design is shown below. Assume that the memory system has 16 address lines denoted by A15 to A0. What is the range of addresses (in hexadecimal) of the memory system that can get enabled by the chip select (CS) signal?

- (A) C800 to CFFF
- (B) CA00 to CAFF
- (C) C800 to C8FF
- (D) DA00 to DFFF

Question 112 of 259 | GATE 2019 Single Session | Q.4 (Computer Science & IT)

Q.4

In 16-bit 2's complement representation, the decimal number -28 is:

- (A) 1111 1111 0001 1100
- (B) 0000 0000 1110 0100
- (C) 1111 1111 1110 0100
- (D) 1000 0000 1110 0100

Question 113 of 259 | GATE 2019 Single Session | Q.5 (Computer Science & IT)

Q.5

Let $U = \{1, 2, \dots, n\}$. Let $A = \{(x, X) | x \in X, X \subseteq U\}$. Consider the following two statements on $|A|$.

I.

$$|A| = n2^{n-1}$$

II.

$$|A| = \sum_{k=1}^n k \binom{n}{k}$$

k(n

k)

n

k=1

Which of the above statements is/are TRUE?

- (A) Only I
- (B) Only II
- (C) Both I and II
- (D) Neither I nor II CS 2/15

Question 114 of 259 | GATE 2019 Single Session | Q.6 (Computer Science & IT)

Q.6

Which one of the following is NOT a valid identity?

- (A) $(x \oplus y) \oplus z = x \oplus (y \oplus z)$
- (B) $(x + y) \oplus z = x \oplus (y + z)$
- (C) $x \oplus y = x + y$, if $xy = 0$
- (D) $x \oplus y = (xy + x'y)'$

Question 115 of 259 | GATE 2019 Single Session | Q.8 (Computer Science & IT)

Q.8

Consider $Z = X - Y$, where X , Y and Z are all in sign-magnitude form. X and Y are each represented in n bits. To avoid overflow, the representation of Z would require a minimum of:

- (A) n bits
- (B) $n-1$ bits
- (C) $n+1$ bits
- (D) $n+2$ bits

Question 116 of 259 | GATE 2019 Single Session | Q.25 (Computer Science & IT)

Q.25 Consider a sequence of 14 elements: $A = [-5, -10, 6, 3, -1, -2, 13, 4, -9, -1, 4, 12, -3, 0]$.

The subsequence sum $S(i, j) = \sum$

$A[k]$

j

$k=i$

. Determine the maximum of $S(i, j)$,

where $0 \leq i \leq j < 14$. (Divide and conquer approach may be used.)

Answer: _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 117 of 259 | GATE 2019 Single Session | Q.37 (Computer Science & IT)

Q.37 There are n unsorted arrays: $A_1, A_2, \dots, A_n$. Assume that n is odd. Each of $A_1, A_2, \dots, A_n$ contains n distinct elements. There are no common elements between any two arrays. The worst-case time complexity of computing the median of the medians of $A_1, A_2, \dots, A_n$ is

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(n^2)$
- (D) $\Omega(n^2 \log n)$

Question 118 of 259 | GATE 2019 Single Session | Q.38 (Computer Science & IT)

Q.38 Let G be any connected, weighted, undirected graph.

I. G has a unique minimum spanning tree, if no two edges of G have the same weight.

II. G has a unique minimum spanning tree, if, for every cut of G , there is a unique minimum-weight edge crossing the cut.

Which of the above two statements is/are TRUE?

- (A) I only

- (B) II only
- (C) Both I and II
- (D) Neither I nor II CS 10/15

Question 119 of 259 | GATE 2019 Single Session | Q.39 (Computer Science & IT)

Q.39 Consider the following snapshot of a system running n concurrent processes. Process i is holding X_i instances of a resource R , $1 \leq i \leq n$. Assume that all instances of R are currently in use. Further, for all i , process i can place a request for at most Y_i additional instances of R while holding the X_i instances it already has. Of the n processes, there are exactly two processes p and q such that $Y_p = Y_q = 0$. Which one of the following conditions guarantees that no other process apart from p and q can complete execution?

- (A) $X_p + X_q < \min \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
- (B) $X_p + X_q < \max \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
- (C) $\min(X_p, X_q) \geq \min \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
- (D) $\min(X_p, X_q) \leq \max \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$

Question 120 of 259 | GATE 2018 Single Session | Q.4 (Computer Science & IT)

Q.4

Let $\oplus$ and $\boxplus$ denote the Exclusive OR and Exclusive NOR operations, respectively.

Which one of the following is NOT CORRECT?

- (A) $P \oplus \overline{Q} = P \boxplus Q$
- (B) $\overline{P} \oplus Q = P \boxplus Q$
- (C) $\overline{P} \oplus \overline{Q} = P \boxplus Q$
- (D) $(P \boxplus \overline{P}) \boxplus Q = (P \boxplus \overline{P}) \oplus \overline{Q}$

Question 121 of 259 | GATE 2018 Single Session | Q.23 (Computer Science & IT)

Q.23 A 32-bit wide main memory unit with a capacity of 1 GB is built using $256M \times 4$ -bit DRAM chips. The number of rows of memory cells in the DRAM chip is 214. The time taken to perform one refresh operation is 50 nanoseconds. The refresh period is 2 milliseconds. The percentage (rounded to the closest integer) of the time available for performing the memory read/write operations in the main memory unit is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 122 of 259 | GATE 2018 Single Session | Q.2 (Computer Science & IT)

Q. 26 – Q. 55 carry two marks each.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 123 of 259 | GATE 2018 Single Session | Q.30 (Computer Science & IT)

Q.30 Let G be a simple undirected graph. Let TD be a depth first search tree of G . Let TB be a breadth first search tree of G . Consider the following statements.

- (I) No edge of G is a cross edge with respect to TD . (A cross edge in G is between two nodes neither of which is an ancestor of the other in TD .)
 (II) For every edge (u,v) of G , if u is at depth i and v is at depth j in TB , then $|i-j| = 1$.

Which of the statements above must necessarily be true?

- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II CS 10/20

Question 124 of 259 | GATE 2018 Single Session | Q.38 (Computer Science & IT)

Q.38 Consider the following parse tree for the expression $a\#b\$c\$d\#e\#f$, involving two binary operators $\$$ and $\#$.

Which one of the following is correct for the given parse tree?

- (A) $\$$ has higher precedence and is left associative; $\#$ is right associative
- (B) $\#$ has higher precedence and is left associative; $\$$ is right associative
- (C) $\$$ has higher precedence and is left associative; $\#$ is left associative
- (D) $\#$ has higher precedence and is right associative; $\$$ is left associative CS 13/20

Question 125 of 259 | GATE 2018 Single Session | Q.45 (Computer Science & IT)

Q.45 Consider the following program written in pseudo-code. Assume that x and y are integers.

```
Count(x,y) {
  if (y != 1){
    if (x != 1) {
      print("*");
      Count(x/2, y);
    }
    else {
      y = y-1;
      Count(1024, y);
    }
  }
}
```

The number of times that the print statement is executed by the call

Count(1024,1024) is ____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 126 of 259 | GATE 2017 Set 1 (CS1) | Q.11 (Computer Science & IT)

Question Number : 11 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 127 of 259 | GATE 2017 Set 1 (CS1) | Q.10 (Computer Science & IT)

Question Number : 10 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 128 of 259 | GATE 2017 Set 1 (CS1) | Q.12 (Computer Science & IT)

Question Number : 12 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 129 of 259 | GATE 2017 Set 1 (CS1) | Q.14 (Computer Science & IT)

Question Number : 14 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 130 of 259 | GATE 2017 Set 1 (CS1) | Q.13 (Computer Science & IT)

Question Number : 13 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 131 of 259 | GATE 2017 Set 1 (CS1) | Q.16 (Computer Science & IT)

Question Number : 16 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 132 of 259 | GATE 2017 Set 1 (CS1) | Q.15 (Computer Science & IT)

Question Number : 15 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 133 of 259 | GATE 2017 Set 1 (CS1) | Q.18 (Computer Science & IT)

Question Number : 18 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 134 of 259 | GATE 2017 Set 1 (CS1) | Q.19 (Computer Science & IT)

Question Number : 19 Correct : 1 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 135 of 259 | GATE 2017 Set 1 (CS1) | Q.20 (Computer Science & IT)**

Question Number : 20 Correct : 1 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 136 of 259 | GATE 2017 Set 1 (CS1) | Q.17 (Computer Science & IT)**

Question Number : 17 Correct : 1 Wrong : -0.33

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 137 of 259 | GATE 2017 Set 1 (CS1) | Q.22 (Computer Science & IT)**

Question Number : 22 Correct : 1 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 138 of 259 | GATE 2017 Set 1 (CS1) | Q.21 (Computer Science & IT)**

Question Number : 21 Correct : 1 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 139 of 259 | GATE 2017 Set 1 (CS1) | Q.24 (Computer Science & IT)**

Question Number : 24 Correct : 1 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 140 of 259 | GATE 2017 Set 1 (CS1) | Q.23 (Computer Science & IT)**

Question Number : 23 Correct : 1 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 141 of 259 | GATE 2017 Set 1 (CS1) | Q.26 (Computer Science & IT)**

Question Number : 26 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 142 of 259 | GATE 2017 Set 1 (CS1) | Q.27 (Computer Science & IT)**

Question Number : 27 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 143 of 259 | GATE 2017 Set 1 (CS1) | Q.25 (Computer Science & IT)

Question Number : 25 Correct : 1 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 144 of 259 | GATE 2017 Set 1 (CS1) | Q.29 (Computer Science & IT)**

Question Number : 29 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 145 of 259 | GATE 2017 Set 1 (CS1) | Q.30 (Computer Science & IT)**

Question Number : 30 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 146 of 259 | GATE 2017 Set 1 (CS1) | Q.31 (Computer Science & IT)**

Question Number : 31 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 147 of 259 | GATE 2017 Set 1 (CS1) | Q.28 (Computer Science & IT)**

Question Number : 28 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 148 of 259 | GATE 2017 Set 1 (CS1) | Q.32 (Computer Science & IT)**

Question Number : 32 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 149 of 259 | GATE 2017 Set 1 (CS1) | Q.33 (Computer Science & IT)**

Question Number : 33 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 150 of 259 | GATE 2017 Set 1 (CS1) | Q.34 (Computer Science & IT)**

Question Number : 34 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 151 of 259 | GATE 2017 Set 1 (CS1) | Q.36 (Computer Science & IT)**

Question Number : 36 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 152 of 259 | GATE 2017 Set 1 (CS1) | Q.3 (Computer Science & IT)

Question Number : 3

5 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 153 of 259 | GATE 2017 Set 1 (CS1) | Q.38 (Computer Science & IT)

Question Number : 38 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 154 of 259 | GATE 2017 Set 1 (CS1) | Q.39 (Computer Science & IT)

Question Number : 39 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 155 of 259 | GATE 2017 Set 1 (CS1) | Q.37 (Computer Science & IT)

Question Number : 37 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 156 of 259 | GATE 2017 Set 1 (CS1) | Q.41 (Computer Science & IT)

Question Number : 41 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 157 of 259 | GATE 2017 Set 1 (CS1) | Q.40 (Computer Science & IT)

Question Number : 40 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 158 of 259 | GATE 2017 Set 1 (CS1) | Q.43 (Computer Science & IT)

Question Number : 43 Correct : 2 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 159 of 259 | GATE 2017 Set 1 (CS1) | Q.42 (Computer Science & IT)

Question Number : 42 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 160 of 259 | GATE 2017 Set 1 (CS1) | Q.45 (Computer Science & IT)

Question Number : 45 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 161 of 259 | GATE 2017 Set 1 (CS1) | Q.44 (Computer Science & IT)**

Question Number : 44 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 162 of 259 | GATE 2017 Set 1 (CS1) | Q.47 (Computer Science & IT)**

Question Number : 47 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 163 of 259 | GATE 2017 Set 1 (CS1) | Q.48 (Computer Science & IT)**

Question Number : 48 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 164 of 259 | GATE 2017 Set 1 (CS1) | Q.46 (Computer Science & IT)**

Question Number : 46 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 165 of 259 | GATE 2017 Set 1 (CS1) | Q.49 (Computer Science & IT)**

Question Number : 49 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 166 of 259 | GATE 2017 Set 1 (CS1) | Q.50 (Computer Science & IT)**

Question Number : 50 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 167 of 259 | GATE 2017 Set 1 (CS1) | Q.52 (Computer Science & IT)**

Question Number : 52 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 168 of 259 | GATE 2017 Set 1 (CS1) | Q.53 (Computer Science & IT)**

Question Number : 53 Correct : 2 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 169 of 259 | GATE 2017 Set 1 (CS1) | Q.51 (Computer Science & IT)

Question Number : 51 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 170 of 259 | GATE 2017 Set 1 (CS1) | Q.54 (Computer Science & IT)**

Question Number : 54 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 171 of 259 | GATE 2017 Set 1 (CS1) | Q.55 (Computer Science & IT)**

Question Number : 55 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 172 of 259 | GATE 2017 Set 1 (CS1) | Q.56 (Computer Science & IT)**

Question Number : 56 Correct : 1 Wrong : -0.33

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 173 of 259 | GATE 2017 Set 1 (CS1) | Q.57 (Computer Science & IT)**

Question Number : 57 Correct : 1 Wrong : -0.33

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 174 of 259 | GATE 2017 Set 1 (CS1) | Q.58 (Computer Science & IT)**

Question Number : 58 Correct : 1 Wrong : -0.33

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 175 of 259 | GATE 2017 Set 1 (CS1) | Q.59 (Computer Science & IT)**

Question Number : 59 Correct : 1 Wrong : -0.33

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 176 of 259 | GATE 2017 Set 1 (CS1) | Q.60 (Computer Science & IT)**

Question Number : 60 Correct : 1 Wrong : -0.33

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 177 of 259 | GATE 2017 Set 1 (CS1) | Q.61 (Computer Science & IT)**

Question Number : 61 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 178 of 259 | GATE 2017 Set 1 (CS1) | Q.62 (Computer Science & IT)

Question Number : 62 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 179 of 259 | GATE 2017 Set 1 (CS1) | Q.63 (Computer Science & IT)

Question Number : 63 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 180 of 259 | GATE 2017 Set 1 (CS1) | Q.64 (Computer Science & IT)

Question Number : 64 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 181 of 259 | GATE 2017 Set 1 (CS1) | Q.65 (Computer Science & IT)

Question Number : 65 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 182 of 259 | GATE 2017 Set 2 (CS2) | Q.11 (Computer Science & IT)

Question Number : 11 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 183 of 259 | GATE 2017 Set 2 (CS2) | Q.12 (Computer Science & IT)

Question Number : 12 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 184 of 259 | GATE 2017 Set 2 (CS2) | Q.13 (Computer Science & IT)

Question Number : 13 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 185 of 259 | GATE 2017 Set 2 (CS2) | Q.10 (Computer Science & IT)

Question Number : 10 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 186 of 259 | GATE 2017 Set 2 (CS2) | Q.15 (Computer Science & IT)

Question Number : 15 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 187 of 259 | GATE 2017 Set 2 (CS2) | Q.14 (Computer Science & IT)

Question Number : 14 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 188 of 259 | GATE 2017 Set 2 (CS2) | Q.17 (Computer Science & IT)

Question Number : 17 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 189 of 259 | GATE 2017 Set 2 (CS2) | Q.18 (Computer Science & IT)

Question Number : 18 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 190 of 259 | GATE 2017 Set 2 (CS2) | Q.16 (Computer Science & IT)

Question Number : 16 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 191 of 259 | GATE 2017 Set 2 (CS2) | Q.20 (Computer Science & IT)

Question Number : 20 Correct : 1 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 192 of 259 | GATE 2017 Set 2 (CS2) | Q.19 (Computer Science & IT)

Question Number : 19 Correct : 1 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 193 of 259 | GATE 2017 Set 2 (CS2) | Q.22 (Computer Science & IT)

Question Number : 22 Correct : 1 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 194 of 259 | GATE 2017 Set 2 (CS2) | Q.23 (Computer Science & IT)

Question Number : 23 Correct : 1 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 195 of 259 | GATE 2017 Set 2 (CS2) | Q.21 (Computer Science & IT)

Question Number : 21 Correct : 1 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 196 of 259 | GATE 2017 Set 2 (CS2) | Q.24 (Computer Science & IT)**

Question Number : 24

Correct : 1 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 197 of 259 | GATE 2017 Set 2 (CS2) | Q.25 (Computer Science & IT)**

Question Number : 25

Correct : 1 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 198 of 259 | GATE 2017 Set 2 (CS2) | Q.26 (Computer Science & IT)**

Question Number : 26

Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 199 of 259 | GATE 2017 Set 2 (CS2) | Q.27 (Computer Science & IT)**

Question Number : 27 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 200 of 259 | GATE 2017 Set 2 (CS2) | Q.28 (Computer Science & IT)**

Question Number : 28 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 201 of 259 | GATE 2017 Set 2 (CS2) | Q.29 (Computer Science & IT)**

Question Number : 29 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 202 of 259 | GATE 2017 Set 2 (CS2) | Q.30 (Computer Science & IT)**

Question Number : 30 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 203 of 259 | GATE 2017 Set 2 (CS2) | Q.32 (Computer Science & IT)**

Question Number : 32 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 204 of 259 | GATE 2017 Set 2 (CS2) | Q.31 (Computer Science & IT)

Question Number : 31 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 205 of 259 | GATE 2017 Set 2 (CS2) | Q.34 (Computer Science & IT)

Question Number : 34 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 206 of 259 | GATE 2017 Set 2 (CS2) | Q.35 (Computer Science & IT)

Question Number : 35 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 207 of 259 | GATE 2017 Set 2 (CS2) | Q.33 (Computer Science & IT)

Question Number : 33 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 208 of 259 | GATE 2017 Set 2 (CS2) | Q.37 (Computer Science & IT)

Question Number : 37 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 209 of 259 | GATE 2017 Set 2 (CS2) | Q.38 (Computer Science & IT)

Question Number : 38 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 210 of 259 | GATE 2017 Set 2 (CS2) | Q.36 (Computer Science & IT)

Question Number : 36 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 211 of 259 | GATE 2017 Set 2 (CS2) | Q.40 (Computer Science & IT)

Question Number : 40 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 212 of 259 | GATE 2017 Set 2 (CS2) | Q.39 (Computer Science & IT)

Question Number :

39 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 213 of 259 | GATE 2017 Set 2 (CS2) | Q.42 (Computer Science & IT)**

Question Number : 42 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 214 of 259 | GATE 2017 Set 2 (CS2) | Q.41 (Computer Science & IT)**

Question Number : 41 Correct : 2 Wrong : -0.66

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 215 of 259 | GATE 2017 Set 2 (CS2) | Q.43 (Computer Science & IT)**

Question Number : 43 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 216 of 259 | GATE 2017 Set 2 (CS2) | Q.45 (Computer Science & IT)**

Question Number : 45 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 217 of 259 | GATE 2017 Set 2 (CS2) | Q.44 (Computer Science & IT)**

Question Number : 44 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 218 of 259 | GATE 2017 Set 2 (CS2) | Q.47 (Computer Science & IT)**

Question Number : 47 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 219 of 259 | GATE 2017 Set 2 (CS2) | Q.46 (Computer Science & IT)**

Question Number : 46 Correct : 2 Wrong : 0

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 220 of 259 | GATE 2017 Set 2 (CS2) | Q.49 (Computer Science & IT)**

Question Number : 49 Correct : 2 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 221 of 259 | GATE 2017 Set 2 (CS2) | Q.50 (Computer Science & IT)

Question Number : 50 Correct : 2 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 222 of 259 | GATE 2017 Set 2 (CS2) | Q.48 (Computer Science & IT)

Question Number : 48 Correct : 2 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 223 of 259 | GATE 2017 Set 2 (CS2) | Q.52 (Computer Science & IT)

Question Number : 52 Correct : 2 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 224 of 259 | GATE 2017 Set 2 (CS2) | Q.53 (Computer Science & IT)

Question Number : 53 Correct : 2 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 225 of 259 | GATE 2017 Set 2 (CS2) | Q.51 (Computer Science & IT)

Question Number : 51 Correct : 2 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 226 of 259 | GATE 2017 Set 2 (CS2) | Q.55 (Computer Science & IT)

Question Number : 55 Correct : 2 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 227 of 259 | GATE 2017 Set 2 (CS2) | Q.54 (Computer Science & IT)

Question Number : 54 Correct : 2 Wrong : 0

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 228 of 259 | GATE 2017 Set 2 (CS2) | Q.56 (Computer Science & IT)

Question Number : 56 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 229 of 259 | GATE 2017 Set 2 (CS2) | Q.57 (Computer Science & IT)

Question Number : 57 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 230 of 259 | GATE 2017 Set 2 (CS2) | Q.58 (Computer Science & IT)

Question Number : 58 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 231 of 259 | GATE 2017 Set 2 (CS2) | Q.59 (Computer Science & IT)

Question Number : 59 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 232 of 259 | GATE 2017 Set 2 (CS2) | Q.61 (Computer Science & IT)

Question Number : 61 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 233 of 259 | GATE 2017 Set 2 (CS2) | Q.62 (Computer Science & IT)

Question Number : 62 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 234 of 259 | GATE 2017 Set 2 (CS2) | Q.63 (Computer Science & IT)

Question Number : 63 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 235 of 259 | GATE 2017 Set 2 (CS2) | Q.64 (Computer Science & IT)

Question Number : 64 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 236 of 259 | GATE 2017 Set 2 (CS2) | Q.60 (Computer Science & IT)

Question Number : 60 Correct : 1 Wrong : -0.33

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 237 of 259 | GATE 2017 Set 2 (CS2) | Q.65 (Computer Science & IT)

Question Number : 65 Correct : 2 Wrong : -0.66

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 238 of 259 | GATE 2016 Set 1 (CS1) | Q.2 (Computer Science & IT)

Q.2

Let a_n be the number of n -bit strings that do NOT contain two consecutive 1s. Which one of the following is the recurrence relation for a_n ?

- (A) $a_n = a_{n-1} + 2a_{n-2}$
- (B) $a_n = a_{n-1} + a_{n-2}$
- (C) $a_n = 2a_{n-1} + a_{n-2}$
- (D) $a_n = 2a_{n-1} + 2a_{n-2}$

Question 239 of 259 | GATE 2016 Set 1 (CS1) | Q.3 (Computer Science & IT)

Q.3

lim

 $x \rightarrow 4$ $\sin(x-4)$ $x-4$

=

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 240 of 259 | GATE 2016 Set 1 (CS1) | Q.6 (Computer Science & IT)

Q.6

Consider the Boolean operator $\#$ with the following properties:

$x\#0 = x$, $x\#1 = \bar{x}$, $x\#x = 0$ and $x\#\bar{x} = 1$. Then $x\#y$ is equivalent to

- (A) $x\bar{y} + \bar{x}y$
- (B) $x\bar{y} + \bar{x}\bar{y}$
- (C) $\bar{x}y + xy$
- (D) $xy + \bar{x}\bar{y}$

Question 241 of 259 | GATE 2016 Set 1 (CS1) | Q.7 (Computer Science & IT)

Q.7

The 16-bit 2's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 242 of 259 | GATE 2016 Set 1 (CS1) | Q.9 (Computer Science & IT)

Q.9

A processor can support a maximum memory of 4 GB, where the memory is word-addressable (a word consists of two bytes). The size of the address bus of the processor is at least bits.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 243 of 259 | GATE 2016 Set 1 (CS1) | Q.11 (Computer Science & IT)

Q.11

Consider the following directed graph:

a
b
c
f
e
d

The number of different topological orderings of the vertices of the graph is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 244 of 259 | GATE 2016 Set 1 (CS1) | Q.13 (Computer Science & IT)

Q.13

The worst case running times of Insertion sort, Merge sort and Quick sort, respectively, are:

- (A) $\Theta(n \log n)$, $\Theta(n \log n)$, and $\Theta(n^2)$
- (B) $\Theta(n^2)$, $\Theta(n^2)$, and $\Theta(n \log n)$
- (C) $\Theta(n^2)$, $\Theta(n \log n)$, and $\Theta(n \log n)$
- (D) $\Theta(n^2)$, $\Theta(n \log n)$, and $\Theta(n^2)$

Question 245 of 259 | GATE 2016 Set 1 (CS1) | Q.19 (Computer Science & IT)

Q.19

Consider the following code segment.

```
x = u - t;  
y = x * v;  
x = y + w;  
y = t - z;  
y = x * y;
```

The minimum number of total variables required to convert the above code segment to static single assignment form is

.

CS(Set A)

6/17

GATE 2016

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 246 of 259 | GATE 2016 Set 1 (CS1) | Q.2 (Computer Science & IT)**

Q. 26 - Q. 55 carry two marks each.

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 247 of 259 | GATE 2016 Set 1 (CS1) | Q.26 (Computer Science & IT)**

Q.26

The coefficient of x^{12} in $(x^3 + x^4 + x^5 + x^6 + \dots)^3$ is

.

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 248 of 259 | GATE 2016 Set 1 (CS1) | Q.40 (Computer Science & IT)**

Q.40

$G = (V, E)$ is an undirected simple graph in which each edge has a distinct weight, and e is a particular edge of G . Which of the following statements about the minimum spanning trees (MSTs) of G is/are TRUE?

- I. If e is the lightest edge of some cycle in G , then every MST of G includes e
- II. If e is the heaviest edge of some cycle in G , then every MST of G excludes e

- (A) I only
- (B) II only
- (C) both I and II
- (D) neither I nor II

Question 249 of 259 | GATE 2016 Set 1 (CS1) | Q.55 (Computer Science & IT)

Q.55

A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1Kbps = 1000 bits/second). Size of an acknowledgement is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds.

Assuming no frame is lost, the sender throughput is
bytes/second.

CS(Set A)

17/17

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 250 of 259 | GATE 2016 Set 2 (CS2) | Q.1 (Computer Science & IT)

Q.1

Consider the following expressions:

- (i) false
- (ii) Q
- (iii) true
- (iv) $P \sqcup Q$
- (v) $\neg Q \sqcup P$

The number of expressions given above that are logically implied by $P \sqcup (P \sqcup Q)$ is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 251 of 259 | GATE 2016 Set 2 (CS2) | Q.8 (Computer Science & IT)

Q.8

Let, $x_1 \sqcup x_2 \sqcup x_3 \sqcup x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\sqcup$ is the XOR operator.

Which one of the following must always be TRUE?

- (A) $x_1 x_2 x_3 x_4 = 0$
- (B) $x_1 x_3 + x_2 = 0$
- (C) $\neg x_1 \sqcup \neg x_3 = \neg x_2 \sqcup \neg x_4$
- (D) $x_1 + x_2 + x_3 + x_4 = 0$

Question 252 of 259 | GATE 2016 Set 2 (CS2) | Q.9 (Computer Science & IT)

Q.9

Let X be the number of distinct 16-bit integers in 2's complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation.Then $X - Y$ is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 253 of 259 | GATE 2016 Set 2 (CS2) | Q.13 (Computer Science & IT)

Q.13

Assume that the algorithms considered here sort the input sequences in ascending order. If the input is already in ascending order, which of the following are TRUE?

- I. Quicksort runs in $\Theta(n^2)$ time
- II. Bubblesort runs in $\Theta(n^2)$ time
- III. Mergesort runs in $\Theta(n)$ time
- IV. Insertion sort runs in $\Theta(n)$ time

- (A) I and II only

- (B) I and III only
- (C) II and IV only
- (D) I and IV only CS(Set B) 3/18 GATE 2016

Question 254 of 259 | GATE 2016 Set 2 (CS2) | Q.23 (Computer Science & IT)

Q.23

Anarkali digitally signs a message and sends it to Salim. Verification of the signature by Salim requires

- (A) Anarkali's public key.
- (B) Salim's public key.
- (C) Salim's private key.
- (D) Anarkali's private key.

Question 255 of 259 | GATE 2016 Set 2 (CS2) | Q.2 (Computer Science & IT)

Q. 26 - Q. 55 carry two marks each.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 256 of 259 | GATE 2016 Set 2 (CS2) | Q.28 (Computer Science & IT)

Q.28

Consider a set U of 23 different compounds in a Chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U . Consider the following statements:

- I. Each compound in $U \setminus S$ reacts with an odd number of compounds.
- II. At least one compound in $U \setminus S$ reacts with an odd number of compounds.
- III. Each compound in $U \setminus S$ reacts with an even number of compounds.

Which one of the above statements is ALWAYS TRUE?

- (A) Only I
- (B) Only II
- (C) Only III
- (D) None

Question 257 of 259 | GATE 2016 Set 2 (CS2) | Q.40 (Computer Science & IT)

Q.40

The number of ways in which the numbers 1, 2, 3, 4, 5, 6, 7 can be inserted in an empty binary search tree, such that the resulting tree has height 6, is

Note: The height of a tree with a single node is 0.

CS(Set B)

11/18

GATE 2016

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 258 of 259 | GATE 2016 Set 2 (CS2) | Q.54 (Computer Science & IT)

Q.54

For the IEEE 802.11 MAC protocol for wireless communication, which of the following statements is/are TRUE?

- I. At least three non-overlapping channels are available for transmissions.
- II. The RTS-CTS mechanism is used for collision detection.
- III. Unicast frames are ACKed.

- (A) All I, II, and III
- (B) I and III only
- (C) II and III only
- (D) II only

Question 259 of 259 | GATE 2016 Set 2 (CS2) | Q.55 (Computer Science & IT)

Q.55

Consider a 128×10^3 bits/second satellite communication link with one way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgement. The minimum number of bits required for the sequence number field to achieve 100% utilization is

.

CS(Set B)

18/18

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]